

The matrix Weyl index in the original phase coordinates

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The preceding lesson constructed the full compact relative projector from both ordered Weyl errors and compared its classical Chern form with the cutoff differential form. Here we identify its formal trace using the exact flat algebra, the original cyclic cocycle and the higher algebraic index theorem. Two orientation signs arise in different places and cancel. We then transfer the resulting matrix formula from the isotropic symbol metric to the original product metric and left quantization.

Read Relative Weyl projectors and the Chern cutoff form for (RP1)–(RP16), (CP1)–(CP14) and (BN1)–(BN5). Read Radial compression and index transport for product-metric symbols for the full RC/PT endpoint argument. The Scaled Weyl parametrices and the surviving differential degree lesson defines the separate finite ordered Weyl coefficient; the present source-theorem route proves its integrated value without deleting any of its terms. The Weyl kernels, operator traces, and a finite trace-class test lesson fixes the original $(2\pi)^{-n}$ operator-trace normalization.

The external theorem used below is the higher algebraic index theorem of M. Pflaum, H. Posthuma and X. Tang. Its degree-zero specialization is stated at the point of use with the actual compact relative pair and full curvature. The local top Riemann–Roch input cited in their proof is due to B. Feigin, G. Felder and B. Shoikhet. Our proof here establishes the exact product map, cocycle value, characteristic side and analytic receiving map for the original course symbols.

1. The full source product and flat connection

1.1 The full product and parameter dictionary

Keep the original phase-space coordinates $z = (x_1, \xi_1, \dots, x_n, \xi_n)$, original orientation $\Omega_{\text{AN}} = \bigwedge_{j=1}^n (dx_j \wedge d\xi_j)$, full matrix product order and formal parameter λ . The full compact-symbol product is:

$$f \#_{\lambda} g = \sum_{k \geq 0} \frac{\lambda^k}{(2i)^k k!} \left[\sum_{j=1}^n (\partial_{\xi_j} \otimes \partial_{x_j} - \partial_{x_j} \otimes \partial_{\xi_j}) \right]^k (f \otimes g) \Big|_{\text{matrix multiplication}}. \quad (\text{FM1})$$

The source writes $\omega = \sum_j dp_j \wedge dq_j$ and

$$u \star_{\hbar} v = \sum_{k \geq 0} \frac{\hbar^k}{2^k k!} \left[\sum_{j=1}^n (\partial_{p_j} \otimes \partial_{q_j} - \partial_{q_j} \otimes \partial_{p_j}) \right]^k (u \otimes v) \Big|_{\text{matrix multiplication}}. \quad (\text{FM2})$$

For smooth compact symbols, (FM2) means the coefficientwise extension of the source's polynomial product: at each fixed k it contains finitely many derivatives, so it is defined without asserting convergence of the infinite $\hbar$ -series.

Define the coordinate swap and coefficientwise parameter substitution

$$F(x, \xi) = (p = \xi, q = x), \quad \lambda = i\hbar, \quad (\Phi f)(p, q; \hbar) = f(q, p; i\hbar). \quad (\text{FM3})$$

This is an isomorphism of Laurent coefficient rings because $i \neq 0$. The source bidifferential operator in (FM2), acting on $\Phi f \otimes \Phi g$, is exactly this course operator in (FM1) after $x = q, \xi = p$. Also $(i\hbar)^k / (2i)^k = \hbar^k / 2^k$ for every nonnegative k . The matrix factors remain in their written order. Thus **each** finite coefficient agrees, including $k = 0$ and every noncommutative derivative word:

$$\Phi(f \#_{\lambda} g) = (\Phi f) \star_{\hbar} (\Phi g). \quad (\text{FM4})$$

The inverse is $(\Phi^{-1}u)(x, \xi; \lambda) = u(\xi, x; \lambda/i)$. Hence (FM4) is an algebra isomorphism of the actual formal smooth matrix-symbol algebras, not merely a comparison of their first Poisson brackets.

The support statement is equally exact. Every term of either product is a local differential expression, so for coefficientwise compact f, g ,

$$\text{supp}(\Phi f) = F(\text{supp } f), \quad \text{supp}(f \#_{\lambda} g) \subseteq \text{supp } f \cap \text{supp } g. \quad (\text{FM5})$$

The first equality uses the inverse linear diffeomorphism and nonzero parameter substitution; the second applies coefficient by coefficient. No completion changes the original support.

1.2 One exact flat Fedosov realization with the negative curvature

The source describes its general Fedosov curvature as the **full** closed formal form $\Omega_F = -\omega + \hbar\omega_1 + \hbar^2\omega_2 + \dots$, *CyclicWeyl.tex* lines 365–376. To see concretely that the product (FM2) has a flat Fedosov realization whose curvature retains the stipulated negative leading term, take the trivial Weyl bundle over the same (p, q) -space. Write P_j, Q_j for fiber coordinates, with the **same** fiber product (FM2), so $[P_j, Q_k]_{\star} = \hbar\delta_{jk}$. Use the flat base connection $d_{p,q}$, zero curvature $\tilde{R} = 0$, and the explicit one-form

$$A_0 = \sum_j (Q_j dp_j - P_j dq_j), \quad B = -2 \sum_j p_j dq_j, \quad A = A_0 + B. \quad (\text{FM6})$$

Here B is central in the fiber Weyl algebra; it changes the chosen curvature representative but contributes nothing to the commutator derivation. For every fiber section U ,

$$DU = d_{p,q}U + \hbar^{-1}[A, U]_{\star} = \sum_j [dp_j(\partial_{p_j} - \partial_{P_j})U + dq_j(\partial_{q_j} - \partial_{Q_j})U]. \quad (\text{FM7})$$

Indeed $[Q_j, U]_{\star} = -\hbar\partial_{P_j}U$ and $[P_j, U]_{\star} = \hbar\partial_{Q_j}U$ exactly: a linear fiber coordinate has no second or higher derivatives. Thus (FM7) has no hidden higher-order term. Its unique horizontal lift with value $u(p, q; \hbar)$ at $P = Q = 0$ is the coefficientwise Taylor section $\hat{u}(p, q; P, Q; \hbar) = u(p + P, q + Q; \hbar)$; direct substitution into (FM7) gives $D\hat{u} = 0$.

The source curvature formula for the flat base connection is $\Omega_F = dA + (2\hbar)^{-1}[A, A]_{\star}$, with the **graded** bracket of one-forms. Since P, Q are fiber variables independent of the base, $dA_0 = 0$. The central B satisfies $dB = -2\omega$ and has no graded commutator contribution. The remaining product is

$$\hbar^{-1}A_0 \wedge_{\star} A_0 = \hbar^{-1} \sum_j ((-Q_j) \star P_j + P_j \star Q_j) dp_j \wedge dq_j = \hbar^{-1} \sum_j [P_j, Q_j]_{\star} dp_j \wedge dq_j = \omega. \quad (\text{FM8})$$

In the first displayed coefficient, $(-Q_j) \star P_j + P_j \star Q_j$ is exactly $[P_j, Q_j]_{\star}$; all cross-index contributions cancel by the separate coordinate commutators. Consequently

$$\Omega_F = -2\omega + \omega = -\omega, \quad D^2 = \hbar^{-1}[\Omega_F, -]_{\star} = 0. \quad (\text{FM9})$$

This explicit representative has all higher $\omega_j = 0$. The full general source curvature remains in (FC4)–(FC7); the central shift stays in the cocycle computation below.

The central term is a choice of lift of the derivation-valued connection, not a degree-zero perturbation of its noncentral Fedosov correction. The source's exact derivation sequence in *CyclicWeyl.tex* lines 37–43 has the constants as kernel. In the present flat model the actual derivation is $d - \delta$ in (FM7), with zero higher noncentral correction. Both A_0 and $A_0 + B$ give this same derivation and hence the same horizontal algebra at every coefficient. The antecedent *FFSrevised.tex* lines 2189–2194 expressly permits a central one-form change of this lift before choosing its Chern–Weil representative. We keep the complete $A_0 + B$, including $dB = -2\omega$, rather than identify the two curvature forms pointwise.

For the particular degree-zero pairing used here, the equality of the trace densities under this central change is also direct. In the full determinant contraction (WR9), every one of the $2n$ connection slots is differentiated once in a fiber coordinate. Any term with B in one such slot vanishes because B is fiber constant. Thus the complete cochain on $A_0 + B$ equals that on A_0 ; no matrix word or noncentral term is removed. On the characteristic side the two closed curvature representatives are $\Omega_0 = +\omega$ and $\Omega_1 = -\omega = \Omega_0 + dB$. For the full compact closed relative Chern form γ of (FC3), put $\Omega_s = \Omega_0 + s dB$. The exact transgression, retaining the full exponential and its inverse parameter, is

$$\frac{d}{ds} [\gamma \wedge e^{-\Omega_s/(2\pi i \hbar)}]_{2n} = d \left[-\frac{1}{2\pi i \hbar} \gamma \wedge B \wedge e^{-\Omega_s/(2\pi i \hbar)} \right]_{2n-1}. \quad (\text{FC8})$$

Indeed $d\gamma = 0 = d\Omega_s$, every degree of γ is even, and dB commutes with the other two-forms; differentiating the finite top-degree exponential gives exactly the displayed factor. The rank-difference term of γ is identically zero, and every remaining coefficient has the original compact support K . The primitive in (FC8) is therefore compactly supported despite the noncompact B . Stokes followed by integration over $s \in [0, 1]$ proves equality of the two characteristic integrals, coefficient by coefficient. Consequently the source pairing on the actual flat connection and the source negative-curvature representative used in (MB3) agree by an explicit cochain calculation and an exact compact transgression. No assumption that the central term belongs to the source's degree-at-least-three noncentral correction is needed.

For horizontal lifts, fiber differentiation of $\hat{u}$ is base differentiation of u . Therefore

$$\hat{u} \star_{\text{fiber}} \hat{v} = \widehat{u \star_h v}, \quad (\hat{u} \star_{\text{fiber}} \hat{v})|_{P=Q=0} = u \star_h v. \quad (\text{FM10})$$

This establishes the source-form Fedosov product for the explicit connection at **every** formal order and shows why (FM4), not an assumed gauge equivalence, is the exact algebra map used below.

1.3 Transport of the actual compact relative projector

Let $e_\infty, e_0 \in M_{2\nu}$ be the specific formal idempotents of RP11, built from the original ordered $a, b = \psi a^{-1}$ and both error sides. RP11 proves

$$e_\infty \#_\lambda e_\infty = e_\infty, \quad e_0 \#_\lambda e_0 = e_0, \quad \text{supp}(e_\infty - e_0)_k \subseteq K := \text{supp}(1 - \psi) \quad \text{for every coefficient } k. \quad (\text{FM11})$$

Apply (FM4) to the **whole** matrix idempotents, not merely to their zeroth terms. With $E_\infty = \Phi e_\infty$ and $E_0 = \Phi e_0$, one obtains

$$E_\infty \star_h E_\infty = E_\infty, \quad E_0 \star_h E_0 = E_0, \quad \text{supp}(E_\infty - E_0)_k \subseteq F(K). \quad (\text{FM12})$$

The reference rank, all original matrix off-diagonal terms, powers of i , and both original error orders survive. The horizontal-lift map of (FM7)–(FM10) realizes E_∞, E_0 as flat Fedosov idempotents for this explicit connection. Their difference is coefficientwise compact in the same fixed set $F(K)$, so every local source density evaluated **on the difference** has a legitimate compact support. This supplies the actual compact relative pair to the source theorem.

The orientation remains separate from the algebra map:

$$F^*(\omega^n/n!) = (-1)^n \Omega_{\text{AN}}, \quad \deg_{\text{or}} F = (-1)^n. \quad (\text{FM13})$$

2. Degree-zero pairing and the exact cyclic sign

Let $\Psi = \Psi_{2n}^{2n}$ be the source trace density of `CyclicWeyl.tex` definition `dfn:psi`. For a compactly supported flat formal symbol f , use the source's own integral functional, as written in `HAnaInThm.tex` lines 541–548:

$$T_\Psi(f) := \int_M \Psi(f). \quad (\text{NS1})$$

The source's definition `dfn:chi`, at $i = r = 0$ and $\alpha_0 = 1$, gives directly

$$\chi_{0,M}^0(1)(f) = \int_M 1 \wedge \Psi_{2n}^{2n}(f) = T_\Psi(f). \quad (\text{NS2})$$

Its subsequent construction, `CyclicWeyl.tex` equation `consmor2`, includes the explicit factor $(2\pi\sqrt{-1})^{-n}$. At degree zero there is one summand, so no cyclic-periodicity convention or higher cochain can change the following exact identity:

$$Q_M^0(1)(f) = \frac{1}{(2\pi i)^n} \chi_{0,M}^0(1)(f) = \frac{1}{(2\pi i)^n} T_\Psi(f). \quad (\text{NS3})$$

This statement uses only the source definitions and is valid for the compact-support algebra on the actual flat $M = \mathbb{R}^{2n}$. The source extends the Weyl cocycles to finite matrices by the ordered ordinary trace, `IndThms.tex` lines 163–171. Thus (NS3) holds with f replaced by a compactly supported matrix symbol and T_Ψ replaced by its matrix-trace extension.

For the relative pair (e_∞, e_0) of the preceding lesson, the difference is coefficientwise supported in the fixed compact $K = \text{supp}(1 - \psi)$. Locality of Ψ keeps its evaluation on the difference supported there. At cyclic degree zero, the source Chern chain is exactly $c_0(e) = e$, `IndThms.tex` lines 21–44; the higher c_j do not enter. By linearity and the source pairing definition at lines 123–134,

$$\langle Q^0(1), e_\infty - e_0 \rangle = \frac{1}{(2\pi i)^n} \int_{\mathbb{R}^{2n}} \Psi^{(2\nu)}(e_\infty - e_0). \quad (\text{NS4})$$

No finite-parametrix error, matrix order, rank reference or support term has been discarded. Equation (NS4) is the exact source cyclic pairing-to-source trace-density morphism needed before a comparison with RP16.

2.1 The operator that the source's proof specifies

Write $z_{2s-1} = P_s, z_{2s} = Q_s$ for $1 \leq s \leq n$. On $2k$ ordered tensor slots $r = 0, \dots, 2k-1$, let $D_{ra} = \partial_{z_a}^{(r)}$. All the D_{ra} commute, even when different partial derivatives act on the same slot. The source's Poisson operator is

$$\alpha_{rs} = \sum_{\ell=1}^n (D_{r,2\ell-1} D_{s,2\ell} - D_{r,2\ell} D_{s,2\ell-1}), \quad \alpha_{sr} = -\alpha_{rs}. \quad (\text{WR1})$$

The plus sign before the displayed minus is ordinary addition; no $1/2$ is included in α . For a k -element subset $I \subset \{1, \dots, n\}$, let D_I be the $2k$ -by- $2k$ matrix with rows the tensor slots and columns P_ℓ, Q_ℓ for $\ell \in I$, in increasing plane order. Define the full alternating operator

$$\Pi_{2k}^{[n]} = \sum_{\substack{I \subset \{1, \dots, n\} \\ |I|=k}} \det D_I, \quad \Pi_0^{[n]} = 1, \quad \Pi_{2k}^{[n]} = 0 \quad (k > n). \quad (\text{WR2})$$

For $k = n$, this is precisely the full determinant with its slots relabelled $0, \dots, 2n-1$, including every one of the $(2n)!$ signed permutations and no factorial multiplier. The superscript $[n]$ records the *ambient* number of symplectic planes; it is not a matrix rank.

Here is an exact operator identity for $0 \leq k < n$. Let $(\Pi_{2k}^{[n]})^i$ act on the $2k$ ordered slots of a $2k+2$ -slot tensor other than slots 0 and i . Then

$$\boxed{\Pi_{2k+2}^{[n]} = \sum_{i=1}^{2k+1} (-1)^i \alpha_{i0} (\Pi_{2k}^{[n]})^i.} \quad (\text{WR3})$$

To prove it without a normalization convention, introduce exterior basis symbols $e_0, \dots, e_{2k+1}$ over the commutative ring generated by the D_{ra} . Put $v_a = \sum_r D_{ra} e_r$. Direct expansion of (WR1) gives

$$\beta := \sum_{r < s} \alpha_{rs} e_r \wedge e_s = \sum_{\ell=1}^n v_{2\ell-1} \wedge v_{2\ell}. \quad (\text{WR4})$$

In $\beta^{k+1}/(k+1)!$, the square of each decomposable plane term vanishes. A set I of $k+1$ *distinct* planes occurs $(k+1)!$ times before division. The coefficient of $e_0 \wedge \dots \wedge e_{2k+1}$ is therefore $\sum_{|I|=k+1} \det D_I = \Pi_{2k+2}^{[n]}$. Equivalently, this coefficient is the Pfaffian of the skew matrix (α_{rs}) , with its fully written normalization

$$\text{Pf}_{2k+2}(\alpha) = \frac{1}{2^{k+1}(k+1)!} \sum_{\sigma \in S_{2k+2}} \text{sgn}(\sigma) \prod_{j=1}^{k+1} \alpha_{\sigma(2j-2), \sigma(2j-1)}. \quad (\text{WR5})$$

This follows directly by expanding β^{k+1} : each pairing appears in $2^{k+1}(k+1)!$ ordered and internally oriented forms. Sorting the pair containing slot 0 gives the Pfaffian Laplace expansion

$$\text{Pf}_{2k+2}(\alpha) = \sum_{i=1}^{2k+1} (-1)^{i+1} \alpha_{0i} \text{Pf}_{2k}(\alpha_{\hat{0}\hat{i}}). \quad (\text{WR6})$$

The removed-slot Pfaffian equals $(\Pi_{2k}^{[n]})^i$ by the same (WR4) calculation on those ordered remaining slots. Since $\alpha_{0i} = -\alpha_{i0}$, (WR6) is exactly (WR3). This proof also shows why repeated choices of one symplectic plane cancel rather than add a hidden $k!$: their exterior square is zero.

Pflaum–Posthuma–Tang explicitly take $(\hbar\alpha)^{\wedge 1} = \hbar\alpha$ in their one-plane display, *CyclicWeyl.tex* lines 124–134. In the proof of $b\tau_{2k} = \tau_{2k+1}$, their lines 232–257 assert, on the actual ordered slots, that

$$\sum_{i=1}^{2k+1} (-1)^i \hbar \alpha_{i0} ((\hbar\alpha)^{\wedge k})^i = (\hbar\alpha)^{\wedge(k+1)}. \quad (\text{WR7})$$

They first write the larger $\sum_s$ expression, then explicitly identify its $s=0$ part with the right side; (WR7) is that identification, not an inferred equality from the theorem alone. Starting from their displayed degree-one operator, (WR7) uniquely determines every subsequent power by induction. Equation (WR3), including the $k=0$ base $\alpha_{01} = \Pi_2^{[n]}$, proves the source-specified power is

$$\boxed{(\hbar\alpha)_{\text{PPT}}^{\wedge k} = \hbar^k \Pi_{2k}^{[n]} \quad \text{on every ordered set of } 2k \text{ slots,} \quad 0 \leq k \leq n.} \quad (\text{WR8})$$

The source’s separate remark at lines 113–115 that its top cocycle is Feigin–Felder–Shoikhet’s up to $(-1)^n$ agrees with (WR8). A convention using the *undivided* exterior power would instead give $k! \Pi_{2k}^{[n]}$, fail (WR7) already at $k=2$, and disagree with that top-cocycle remark. The factor of two from an undivided exterior square at $k=2$ is thus resolved by the source’s own recursion.

2.2 Exact value of the source cochain for the written connection

The source definition at CyclicWeyl.tex lines 95–112 and (WR8), with the full ordered exponential product $\mathcal{E}$, give an equality of cochains in every dimension:

$$\tau_{2n}^{\text{PPT}} = (-1)^n \hbar^n \mu_{2n} \int_{\Delta^{2n}} \mathcal{E}(u) \Pi_{2n}^{[n]} du_1 \cdots du_{2n} = \tau_{2n}^{\text{det}}. \quad (\text{WR9})$$

No $n!$ is omitted: (WR5) and (WR8) specify it. For the full FM6 connection

$$A = \sum_{j=1}^n (Q_j dp_j - P_j dq_j) - 2 \sum_{j=1}^n p_j dq_j, \quad D = d - \sum_j (dp_j \partial_{P_j} + dq_j \partial_{Q_j}), \quad \Omega_F = -\omega, \quad (\text{WR10})$$

and each compact formal matrix symbol u , its horizontal section is $\hat{u}(p, q; P, Q) = u(p + P, q + Q)$. The central shift in A supplies the written curvature and has zero fiber derivative; it is retained, not removed from the connection.

At degree $2n$, the source shuffle in CyclicWeyl.tex lines 393–411 has a length-zero first chain and only the identity $(0, 2n)$ -shuffle. Its definition of Ψ at lines 550–570 therefore gives

$$\Psi_{2n}^{2n}(\hat{u}) = \hbar^{-2n} \tau_{2n}^{\text{PPT}}(\hat{u}, A, \dots, A). \quad (\text{WR11})$$

All $2n$ inverse powers, the ordinary ordered matrix trace and the full A are present. By (WR9), this is the source cochain on the written connection. The determinant contraction can be calculated without a normalization shortcut:

Write

$$\beta_{2j-1} := \partial_{P_j} A = -dq_j, \quad \beta_{2j} := \partial_{Q_j} A = dp_j. \quad (\text{FD5})$$

Each factor A is differentiated exactly once in every summand of (the full determinant). Exterior multiplication of the resulting differential-form slots gives

$$\begin{aligned} \bigwedge \Pi_{2n}(\hat{u} \otimes A^{\otimes 2n}) &= \hat{u} \sum_{\sigma \in S_{2n}} \text{sgn}(\sigma) \beta_{\sigma(1)} \wedge \cdots \wedge \beta_{\sigma(2n)} \\ &= (2n)! \hat{u} \beta_1 \wedge \cdots \wedge \beta_{2n} \\ &= (2n)! \hat{u} \bigwedge_{j=1}^n (dp_j \wedge dq_j) = (2n)! \hat{u} \frac{\omega^n}{n!}. \end{aligned} \quad (\text{FD6})$$

The second equality uses the sign of the permutation once from the determinant and once from exterior permutation, so every one of the $(2n)!$ summands has the same sign. The third equality retains each original ordered coordinate pair: $(-dq_j) \wedge dp_j = dp_j \wedge dq_j$.

After (the full determinant) acts, every A -slot has fiber-constant coefficients β_j . Every positive-degree term in any exponential of (the ordered exponential) differentiates one such constant slot and vanishes, including the terms involving slot zero. Thus $\mathcal{E}$ acts as the identity **on this exact tensor**. The simplex volume is $1/(2n)!$, and μ_{2n} evaluates $\hat{u}$ at $P = Q = 0$, yielding u .

These equations retain all $(2n)!$ permutations, the ordered simplex volume $1/(2n)!$, and the vanishing of every positive exponential term after the A -slots become fiber-constant. Hence, coefficientwise for every finite matrix rank m ,

$$\boxed{\Psi_{2n}^{2n, (m)}(\hat{u}) = (-1)^n \hbar^{-n} \text{tr}_m(u) \frac{\omega^n}{n!}.} \quad (\text{WR12})$$

There is no assumption that u is scalar, idempotent or only a zeroth-order symbol. At each formal order the source expression is local in finitely many derivatives, so a fixed coefficientwise compact support remains compact.

Retain the source normalization $(2\pi i)^{-n}$, the exact algebra map $p = \xi, q = x, \lambda = i\hbar$ of FM3–FM4, and both separate orientation signs of FM13. Their product is $+1$ in the phase-space integral. Thus for every compact formal finite-matrix symbol f in the original ordered Moyal algebra,

$$\boxed{\mathbf{Q}^0(1)(\widehat{\Phi}f) = (2\pi i)^{-n} \int_{\mathbb{R}_{p,q}^{2n}} \Psi_{2n}^{2n,(m)}(\widehat{\Phi}f) = (-1)^n (2\pi\lambda)^{-n} \int_{\mathbb{R}_{x,\xi}^{2n}} \text{tr}_m(f) \Omega_{\text{AN}} = (-1)^n \mathcal{T}_\lambda(f).} \quad (\text{WR13})$$

The expression $\mathbf{Q}^0(1)(\widehat{\Phi}f)$ denotes evaluation on the horizontal Fedosov section; it does not pretend that a base function is already in the source's section domain. In particular, for the actual relative idempotent pair,

$$\mathbf{Q}^0(1)(\widehat{\Phi}e_\infty - \widehat{\Phi}e_0) = (-1)^n \mathcal{T}_\lambda(e_\infty - e_0). \quad (\text{WR14})$$

The difference has the fixed compact coefficientwise support proved in the preceding lesson and FM12, so both sides are defined. For $n = 1$, (WR13) is exactly the one-plane negative value; for $n = 2$ it is positive, and for $n = 3$ negative. This parity is proved by (WR3)–(WR12), not extrapolated from three examples.

3. The complete curvature and Chern orientation

Let $P \in C^\infty(\mathbb{R}^{2n}; M_{2\nu}(\mathbb{C}))$ be CP1. The constant reference projector is $e_0 = \text{diag}(0, I_\nu)$. CP1 gives

$$P^2 = P, \quad e_0^2 = e_0, \quad P = e_0 \text{ outside } K = \text{supp}(1 - \psi), \quad \text{rank } P = \text{rank } e_0 = \nu. \quad (\text{FC1})$$

The last rank equality follows from the exact global conjugation $P = U_0 p U_0^{-1}$ in CP1, including where a is noninvertible. Thus the image bundles V_P, V_0 agree by the identity outside K , not merely by an unspecified exterior isomorphism.

Differentiate $P^2 = P$. Multiplying by P on both sides gives $P(dP)P = 0$; replacing P by $I - P$ gives $(I - P)(dP)(I - P) = 0$. Therefore dP is off-diagonal for the splitting $\text{im } P \oplus \ker P$, and every odd power $(dP)^{2k+1}$ has zero full matrix trace. Since $d(dP) = 0$, direct exterior differentiation yields

$$d \text{tr}_{2\nu}[P(dP)^{2k}] = \text{tr}_{2\nu}[(dP)^{2k+1}] = 0 \quad (k \geq 0). \quad (\text{FC2})$$

For $k \geq 1$, these forms vanish outside K because $dP = 0$ there. At $k = 0$, $\text{tr}(P - e_0) = \nu - \nu = 0$ identically. Hence the relative Chern-character representative in whatever fixed source convention is a finite sum of compactly supported closed even forms

$$\gamma = \text{Ch}(V_P - V_0) = \sum_{k=1}^n \gamma_{2k}, \quad \gamma_{2k} = c_k \text{tr}_{2\nu}[P(dP)^{2k}], \quad d\gamma_{2k} = 0, \quad \text{supp } \gamma_{2k} \subseteq K. \quad (\text{FC3})$$

Here c_k records the source Chern normalization, calculated exactly in (CN1)–(CN2). Equation (FC2) proves the needed closure independently of that normalization. The source's theorem uses precisely the difference of bundles determined by the zeroth idempotents, which are V_P, V_0 here.

Keep the source's entire closed formal Fedosov two-form. Its leading term is **negative** the source symplectic form (CyclicWeyl.tex lines 365–376):

$$\Omega_F = -\omega + \hbar\Omega_1 + \hbar^2\Omega_2 + \cdots, \quad d\Omega_F = 0 \quad (\text{FC4})$$

on $\mathbb{R}^{2n}$. No term has been set to zero, even though the base is flat. Each coefficient has an explicit global primitive by the radial Poincaré homotopy: for a closed two-form β , let

$$(H\beta)_z(v) = \int_0^1 s \beta_{sz}(z, v) ds, \quad \eta_F = \sum_{j \geq 0} \hbar^j H\Omega_j \quad (\Omega_0 = -\omega). \quad (\text{FC5})$$

Differentiate $s^2 \beta_{sz}(v, w)$ in s , use the coordinate closedness equation for β , and integrate from zero to one. The zero endpoint is zero, while the other is $\beta_z(v, w)$; this proves $d(H\beta) = \beta$ and hence $d\eta_F = \Omega_F$ coefficient by coefficient. The primitive need not decay, but its product with a compactly supported γ_{2k} is compactly supported.

For every $j \geq 1$, closedness of γ_{2k} and Ω_F gives the exact compact-support identity

$$\gamma_{2k} \wedge \Omega_F^j = d(\gamma_{2k} \wedge \eta_F \wedge \Omega_F^{j-1}). \quad (\text{FC6})$$

The sign is $+1$ because $\deg \gamma_{2k} = 2k$ is even. The primitive is compactly supported by (FC3), so its integral over the original $\mathbb{R}^{2n}$ orientation is zero by Stokes. This handles every formal $\hbar$ -coefficient and every negative $\hbar$ -power from the source exponential without absorbing or dropping the original ω , the corrections Ω_j , or their constants.

Choose the exact flat Euclidean connection on $T\mathbb{R}^{2n}$. Its curvature is zero, so its $\hat{A}$ -form is 1; the source characteristic class is unchanged by this connection choice. In the source theorem's $k = 0, \alpha_0 = 1$ characteristic expression, expand the complete exponential and select top degree. Every term containing a positive power of Ω_F integrates to zero by (FC6), and the rank-difference term is zero by (FC1). Thus the exact reduction is

$$\int_{\mathbb{R}^{2n}} [\hat{A}(T\mathbb{R}^{2n}) \text{Ch}(V_P - V_0) \exp(-\Omega_F/(2\pi i \hbar))]_{2n} = \int_{\mathbb{R}^{2n}} \gamma_{2n}. \quad (\text{FC7})$$

IndThms.tex lines 385–387 defines the invariant Chern polynomial on matrices as $\text{tr}(\exp X)$. Its source pairing formula at lines 431–475 has the global factor $(2\pi i)^{-n}$ and applies this polynomial to the vector-bundle curvature and $-\Omega_F/\hbar$. In the flat tangent case, where the chosen exact Euclidean connection has $\hat{A} = 1$, distributing that factor among the homogeneous terms gives the *entire* top-degree source characteristic integrand

$$\frac{1}{(2\pi i)^n} \sum_{k=0}^n \frac{1}{k!(n-k)!} \text{tr}_{V_P - V_0}(F^k) \wedge \left(-\frac{\Omega_F}{\hbar}\right)^{n-k}. \quad (\text{CN1})$$

Here the $k = 0$ trace means the rank difference, F_P, F_0 are the two actual bundle curvatures, and the displayed polynomial is finite by degree. In particular, the source's degree- $2k$ Chern normalization is

$$\gamma_{2k} = \frac{1}{k!(2\pi i)^k} (\text{tr}_{V_P} F_P^k - \text{tr}_{V_0} F_0^k). \quad (\text{CN2})$$

Indeed the k -th factor $(2\pi i)^{-k}$ in (CN2) times the remaining $(2\pi i)^{-(n-k)}$ in $\exp[-\Omega_F/(2\pi i \hbar)]$ reconstructs **exactly** the global $(2\pi i)^{-n}$ in (CN1). No $\hbar$, 2π , factorial or source curvature sign has been absorbed.

Choose the Grassmann connections allowed by IndThms.tex lines 150–154:

$$\nabla_P s = P ds \quad (Ps = s), \quad \nabla_0 s = e_0 ds \quad (e_0 s = s). \quad (\text{CN3})$$

They agree under the identity outside $K = \text{supp}(1 - \psi)$, because $P = e_0$ there. Differentiate $P^2 = P$ and multiply on both sides to get $P(dP)P = 0$; likewise $(I - P)(dP)(I - P) = 0$. Thus dP is off-diagonal and $(dP)^2$ commutes with P . For a section $s = Ps$,

$$\begin{aligned} \nabla_P^2 s &= P d(P ds) = P(dP) \wedge ds \\ &= P(dP) \wedge [(dP)s + P ds] = P(dP)^2 s. \end{aligned} \quad (\text{CN4})$$

Consequently $F_P = P(dP)^2 P = P(dP)^2$ on V_P , $F_0 = 0$, and for every $k \geq 1$

$$\mathrm{tr}_{V_P}(F_P^k) = \mathrm{tr}_{2\nu}(P(dP)^{2k}). \quad (\text{CN5})$$

The identity follows by multiplying the commuting factors P and $(dP)^2$; it keeps the full matrix order and all $2k$ one-form factors. The traces vanish outside K because $dP = 0$ there. The rank difference at $k = 0$ is zero by CP1.

Equation (FC6) proves that for $k < n$, every term of (CN1) with a positive Ω_F -power has a compactly supported primitive and integrates to zero, coefficient by coefficient in $\hbar$. Substituting (CN5) in the surviving $k = n$ term yields, on the source phase space M_{src} oriented by $\omega^n/n!$, the exact value

$$\int_{M_{\text{src}}}^{\omega} [\hat{A} \mathrm{Ch}(V_P - V_0) \exp(-\Omega_F/(2\pi i \hbar))]_{2n} = \frac{1}{n!(2\pi i)^n} \int_{M_{\text{src}}}^{\omega} \mathrm{tr}_{2\nu}(P_{\text{src}}(dP_{\text{src}})^{2n}). \quad (\text{CN6})$$

Now apply the **same** coordinate map $F(x, \xi) = (p = \xi, q = x)$ as (FM3). Its degree from the original orientation $\Omega = dx_1 \wedge d\xi_1 \wedge \cdots \wedge dx_n \wedge d\xi_n$ to the source symplectic orientation is $(-1)^n$. The source projector is $P_{\text{src}} = P_{\text{AN03}} \circ F^{-1}$, so naturality of d and matrix trace gives $F^* \mathrm{tr}[P_{\text{src}}(dP_{\text{src}})^{2n}] = \mathrm{tr}[P_{\text{AN03}}(dP_{\text{AN03}})^{2n}]$. Unlike the source trace-density density $f \omega^n/n!$, this Chern form supplies **no second standalone volume-form sign**. The oriented change of variables is therefore

$$\begin{aligned} \int_{M_{\text{src}}}^{\omega} [\hat{A} \mathrm{Ch}(V_P - V_0) e^{-\Omega_F/(2\pi i \hbar)}]_{2n} &= \frac{(-1)^n}{n!(2\pi i)^n} \int_{M_{\text{AN03}}}^{\Omega} \mathrm{tr}_{2\nu}(P(dP)^{2n}) \\ &= (-1)^n (2\pi)^{-n} \frac{2}{i^n n!} \int_{M_{\text{AN03}}}^{\Omega} (1 - \psi)^n \mathrm{tr}_{\nu}[(db \wedge da)^n]. \end{aligned} \quad (\text{CN7})$$

4. Applying the theorem to the actual relative projector

4.1 The actual algebra, projectors and source theorem hypotheses

Keep the original isotropic metric $g_z = h(z)(|dx|^2 + |d\xi|^2)$, $h(z) = (1 + |z|^2)^{-1}$, and the original full $a \in S(1, g; \mathrm{End} \mathbb{C}^\nu)$, uniformly invertible outside a compact set. Let ψ be the written exterior cutoff, $b = \psi a^{-1}$ on the invertibility region extended by zero, $t = 1 - \psi$, and $\Omega_{\text{AN}} = dx_1 \wedge d\xi_1 \wedge \cdots \wedge dx_n \wedge d\xi_n$. These are the objects of IP1–IP3, not a normalized replacement. The preceding lesson constructs from both ordered errors the full formal idempotents $e_\infty, e_0 \in M_{2\nu}(C^\infty(\mathbb{R}_{x,\xi}^{2n})[[\lambda]])$ with

$$e_\infty \#_\lambda e_\infty = e_\infty, \quad e_0 \#_\lambda e_0 = e_0, \quad \mathrm{supp}[\lambda^j](e_\infty - e_0) \subseteq K := \mathrm{supp}(1 - \psi) \quad (j \geq 0). \quad (\text{MB1})$$

The individual projectors need not be compactly supported. The coefficientwise support statement is the precise hypothesis needed for the source's relative pairing.

Equations (FM1)–(FM12) prove an algebra identity, not a formal equivalence claim: with $F(x, \xi) = (p = \xi, q = x)$ and $\lambda = i\hbar$, the complete ordered $\#_\lambda$ product becomes the source's written Weyl product $\star_\hbar$ at every coefficient. The explicit FM6 Fedosov one-form, its derivation, curvature and horizontal lift are

$$A = \sum_{j=1}^n (Q_j dp_j - P_j dq_j) - 2 \sum_{j=1}^n p_j dq_j, \quad D = d - \sum_{j=1}^n (dp_j \partial_{P_j} + dq_j \partial_{Q_j}), \quad \Omega_F = -\omega, \quad \widehat{u}(p, q; P, Q) = u(p + P, q + Q). \quad (\text{MB2})$$

The central term in A is essential to the displayed curvature. The two horizontal source projectors $\widehat{E}_\infty = \widehat{\Phi} e_\infty$ and $\widehat{E}_0 = \widehat{\Phi} e_0$ remain idempotent; their difference is coefficientwise supported in $F(K)$ as a section because every Taylor coefficient differentiates a compactly supported base coefficient. Both lie in the source formal Laurent algebra $\mathcal{A}^{((\hbar))}$; in fact they have no negative $\hbar$ -powers before the trace normalization.

At cyclic degree zero take the source theorem's closed total-form sequence $\alpha = (\alpha_0) = (1)$. Pflaum–Posthuma–Tang's theorem `thm:higher-algind`, `IndThms.tex` lines 412–424, applies to these **two actual projectors** with compact difference and states

$$\langle Q^0(1), \widehat{E}_\infty - \widehat{E}_0 \rangle = \int_{\mathbb{R}_{p,q}^{2n}}^\omega \left[\widehat{A}(T\mathbb{R}^{2n}) \operatorname{Ch}(V_P - V_0) \exp\left(-\frac{\Omega_F}{2\pi i \hbar}\right) \right]_{2n}. \quad (\text{MB3})$$

Here V_P, V_0 are determined by the **zeroth** source projectors, not by an exterior-only rank surrogate. Equation (CP1) proves $P = [\lambda^0]e_\infty = U_0 p U_0^{-1}$ globally, $e_0 = \operatorname{diag}(0, I_\nu)$, $\operatorname{rank} P = \operatorname{rank} e_0 = \nu$, and $P = e_0$ outside K . Thus the theorem's bundle and support hypotheses hold. The base is an ordinary symplectic manifold, so no orbifold specialization or unstated properness argument is needed; the source theorem itself expressly assumes compact projector difference on M .

4.2 Both sides in the original coordinates, with every sign

Equations (NS3)–(NS4) identify the source degree-zero pairing with $(2\pi i)^{-n} \int \Psi_{2n}^{2n}$ on a compact relative difference. Equations (WR1)–(WR14) prove the original source wedge normalization from its displayed recurrence and evaluate that *same* Ψ -shuffle for the full A of (MB2). Its all-dimensional identity on the actual projectors is

$$\langle Q^0(1), \widehat{E}_\infty - \widehat{E}_0 \rangle = (-1)^n \mathcal{T}_\lambda(e_\infty - e_0), \quad \mathcal{T}_\lambda(f) = (2\pi\lambda)^{-n} \int_{\mathbb{R}_{x,\xi}^{2n}} \operatorname{tr}_{2\nu}(f) \Omega_{\text{AN}}. \quad (\text{MB4})$$

The two signs in the coordinate-swapped trace-density integral cancel separately: $F^*(\omega^n/n!) = (-1)^n \Omega_{\text{AN}}$ and $\deg_{\text{or}} F = (-1)^n$. The remaining $(-1)^n$ is the source cocycle sign. Every coefficient of the difference has fixed compact support, so (MB4) is an equality in $\mathbb{C}((\lambda))$, not an integral of the individual noncompact projectors.

On the characteristic side, choose the exact flat Euclidean connection for $T\mathbb{R}^{2n}$, giving $\widehat{A} = 1$. Keep the **full** closed source curvature $\Omega_F = -\omega + \hbar\Omega_1 + \cdots$; for the particular connection (MB2), all later coefficients happen to vanish, but Equations (FC5)–(FC6) prove the stronger exact statement needed here: every positive power of Ω_F multiplied by a compactly supported closed positive-degree relative Chern form has a compactly supported primitive and integrates to zero, coefficient by coefficient, including its negative $\hbar$ -powers. The rank-difference 0-form is exactly zero. Therefore FC7 reduces the entire exponential, without suppressing it from (MB3), to the top source Chern form.

Equations (CN1)–(CN6) obtain that form from the source's complete $(2\pi i)^{-n}$ factor and the actual Grassmann curvature $F_P = P(dP)^2$:

$$\int_{\mathbb{R}_{p,q}^{2n}}^\omega [\widehat{A} \operatorname{Ch}(V_P - V_0) e^{-\Omega_F/(2\pi i \hbar)}]_{2n} = \frac{1}{n!(2\pi i)^n} \int_{\mathbb{R}_{p,q}^{2n}}^\omega \operatorname{tr}_{2\nu}[P_{\text{src}}(dP_{\text{src}})^{2n}]. \quad (\text{MB5})$$

In contrast to the trace density, this Chern form has no separate symplectic-volume factor. Naturality gives $F^* \operatorname{tr}[P_{\text{src}}(dP_{\text{src}})^{2n}] = \operatorname{tr}[P(dP)^{2n}]$, while the oriented-domain change contributes one $(-1)^n$. Equations (CN7) and (CP14) then give, with the original cutoff and all matrix words retained,

$$\begin{aligned} \int_{\mathbb{R}_{p,q}^{2n}}^\omega [\widehat{A} \operatorname{Ch}(V_P - V_0) e^{-\Omega_F/(2\pi i \hbar)}]_{2n} &= (-1)^n \frac{1}{n!(2\pi i)^n} \int_{\mathbb{R}_{x,\xi}^{2n}}^{\Omega_{\text{AN}}} \operatorname{tr}_{2\nu}[P(dP)^{2n}] \\ &= (-1)^n (2\pi)^{-n} \frac{2}{i^n n!} \int_{\mathbb{R}_{x,\xi}^{2n}}^{\Omega_{\text{AN}}} (1 - \psi)^n \operatorname{tr}_\nu[(db \wedge da)^n]. \end{aligned} \quad (\text{MB6})$$

The one Chern orientation sign is the same parity as the one source-cocycle sign in (MB4); neither sign has been set to +1 individually.

Combining the two **evaluations of the same source pairing** (MB3)–(MB6) cancels those two factors $(-1)^n$, which are invertible over $\mathbb{C}((\lambda))$. The exact resulting formal identity is

$$\mathcal{T}_\lambda(e_\infty - e_0) = \frac{1}{n!(2\pi i)^n} \int_{\mathbb{R}^{2n}}^{\Omega_{\text{AN}}} \text{tr}_{2\nu}[P(dP)^{2n}] = (2\pi)^{-n} \frac{2}{i^n n!} \int_{\mathbb{R}^{2n}}^{\Omega_{\text{AN}}} (1 - \psi)^n \text{tr}_\nu[(db \wedge da)^n]. \quad (\text{MB7})$$

The right side has no λ : the rank term is zero and FC6 kills every curvature power after integration. This is an equality of the original full formal trace with the exact classical relative form, not only an equality of their leading symbols.

4.3 The analytic index and the further coefficient consequence

Equation (RP16) proves from the original two error orders and the actual analytic Fredholm index that

$$\text{ind } a^w = (2\pi)^{-n} \int_{\mathbb{R}^{2n}}^{\Omega_{\text{AN}}} \text{tr}_{2\nu}[\lambda^n](e_\infty - e_0). \quad (\text{MB8})$$

Taking $[\lambda^0]$ of the **full** left side of (MB7) reproduces precisely (MB8), since $(2\pi\lambda)^{-n}$ shifts the coefficient by n . The right side of (MB7) is constant. Thus the original matrix analytic index formula is proved for every $n \geq 1$ under the stated $S(1, g)$ and exterior invertibility hypotheses:

$$\text{ind } a^w = (2\pi)^{-n} \frac{2}{i^n n!} \int_{\mathbb{R}^{2n}}^{\Omega_{\text{AN}}} (1 - \psi)^n \text{tr}_\nu[(db \wedge da)^n]. \quad (\text{MB9})$$

The preceding lesson proves the complete scalar cutoff primitive, its exact beta moment $(n!)^2/(2n)!$, and Stokes with the original outward boundary orientation. Applying that already proved identity to the *written right side* of (MB9), without a convention change, gives the original boundary coefficient

$$\text{ind } a^w = -(-2\pi i)^{-n} \frac{(n-1)!}{(2n-1)!} \int_{\partial B}^{\Omega_{\text{AN}}} \text{tr}_\nu[(a^{-1} da)^{2n-1}]. \quad (\text{MB10})$$

The boundary lies in the exterior invertibility region and carries the orientation induced by Ω_{AN} . No rank, cutoff term, matrix order, factorial, power of i , or coordinate sign has been discarded.

There is a further exact consequence of (MB7). Write $e_\infty - e_0 = \sum_{j \geq 0} \lambda^j d_j$, each d_j supported in K . Since the full Laurent series $\mathcal{T}_\lambda(e_\infty - e_0)$ is constant, coefficient comparison in $\mathbb{C}((\lambda))$ gives

$$\int_{\mathbb{R}^{2n}}^{\Omega_{\text{AN}}} \text{tr}_{2\nu} d_j = 0 \quad (j \geq 0, j \neq n), \quad (2\pi)^{-n} \int_{\mathbb{R}^{2n}}^{\Omega_{\text{AN}}} \text{tr}_{2\nu} d_n = \text{ind } a^w. \quad (\text{MB11})$$

RP16 had independently established the vanishing for $j < n$; (MB11) supplies every $j > n$ through the source theorem. The conclusion concerns **integrated matrix traces**, not pointwise vanishing of those coefficients.

There is also an analytic proof of every coefficient assertion in (MB11), with no algebraic index theorem used for this part. Fix any integer $L \geq n + 1$, and choose the original finite correction integer $N \geq L + 1$. Repeat (RP12)–(RP15) through degree $L - 1$: outside K , both errors have zero zeroth coefficient, so $C_N - B_\infty$ vanishes through degree $N - 1$. The same full triangular interpolation is idempotent,

and each coefficient of its derivative through degree $L - 1$ is compactly supported. The complete double-commutator in RP14 then has integrated trace zero by RP13 at each of these degrees. This proves equality between the integrated relative-projector coefficient d_j and the coefficient of the two original ordered error powers $r^{\#2N} - s^{\#2N}$ for every $j < L$.

Use the original scaling $a_\varepsilon(z) = a(\varepsilon z)$, its full cutoff inverse, and the original scaled metric g_ε . The induction in DE5–DE7, with its entire high-degree product case, expands those same powers through degree $L - 1$. Since $2N \geq L$, the remainder is uniformly in $S(h_\varepsilon^L, g_\varepsilon)$; coefficients through degree $L - 1$ have the original compact support after the full dilation. IP22–IP28 prove its actual trace norm is $O(\varepsilon^{2L-2n})$, including both factors and all output domains. CI12 retains the full phase Jacobian ε^{-2n} and Fourier factor $(2\pi)^{-n}$. RP6 and the positive-scale index equality thus give, with the same d_j as (MB11),

$$\text{ind } a^w = (2\pi)^{-n} \sum_{j=0}^{L-1} \varepsilon^{2j-2n} \int_{\mathbb{R}^{2n}}^{\Omega_{\text{AN}}} \text{tr}_{2\nu} d_j + O(\varepsilon^{2L-2n}). \quad (\text{AT1})$$

The error constant may depend on the fixed L, N , which is sufficient for the following exact coefficient argument. Start with $j = 0$, multiply by ε^{2n} , and let $\varepsilon \downarrow 0$; this gives the zero coefficient. After each zero coefficient, divide by the next displayed power and repeat, obtaining zero for every $j < n$. At $j = n$ the same limit gives the index. Subtract that exact constant. Divide successively by ε^{2j-2n} for $n < j < L$; every later term and the remainder then tend to zero, so each such coefficient is zero. For an arbitrary prescribed j , choose $L > \max(j, n)$ before this calculation. This proves all of (MB11) analytically for the original full compact projector. The theorem identifies its surviving value with the complete Chern form in (MB7). The independent direct calculation is proved in From ordered Weyl errors to the exterior index form, Sections 2–6, equations (5)–(18), using these same original symbols and errors.

FM6–FM9 and FC8 prove the exact central connection and curvature maps in the figure. RP12–RP15, IP22–IP28 and AT1 prove the displayed analytic coefficient comparison, with every original cutoff, ordered error power, Jacobian and Fourier factor retained. The source connection framework is Pflaum–Posthuma–Tang, *CyclicWeyl.tex* lines 336–382; the central-lift observation is Feigin–Felder–Shoikhet, *FFSrevised.tex* lines 2189–2194.

The scaled Weyl lesson proves $\text{ind } a^w = A_n$, where A_n is its complete finite ordered Weyl coefficient, including every C_k and cutoff derivative. Equation (MB9) identifies its integrated value with the original cutoff expression. This route uses the full relative projector, exact source cyclic trace and source index theorem. The independent direct proof is From ordered Weyl errors to the exterior index form, equations (5)–(18): it proves the complete multi-degree selection, atomic derivative involution, ordered exterior count and full beta moment for every integer error power $N > n$. Its equation (17) is exactly the cutoff expression in (MB9); its equation (18) gives the same outward boundary value in (MB10). The two complete arguments therefore agree on the actual original operators.

5. The product metric and left quantization

5.1 Retain the original product structure

Let $n \geq 1$ and keep the two coordinate groups $z = (x, \xi)$, $x, \xi \in \mathbb{R}^n$, with the actual metrics

$$G_z = \frac{|dx|^2}{\langle x \rangle^2} + \frac{|d\xi|^2}{\langle \xi \rangle^2}, \quad g_z = \frac{|dx|^2 + |d\xi|^2}{1 + |x|^2 + |\xi|^2}, \quad \Omega_{\text{AN}} = dx_1 \wedge d\xi_1 \wedge \cdots \wedge dx_n \wedge d\xi_n. \quad (\text{PM1})$$

Assume the full matrix symbol $a \in S(1, G; \text{End } \mathbb{C}^\nu)$ has a uniformly bounded inverse outside some ball B_{R_0} . Choose the original *cutoff inverse* χ with $\chi = 1$ for $|z| \geq R_0$, supported in the invertibility region,

The complete connection lift and the original analytic coefficients

$$A_0 = \sum_j (Q_j dp_j - P_j dq_j), \quad B = -2 \sum_j p_j dq_j, \quad A = A_0 + B$$

$$d + \hbar^{-1}[A_0, -] = d - \delta \quad \longleftrightarrow \quad d + \hbar^{-1}[A_0 + B, -] = d - \delta$$

B is fiber central: the actual derivation and horizontal algebra are identical.

$$\Omega_0 = +\omega, \quad dB = -2\omega, \quad \Omega_s = \Omega_0 + s dB, \quad \Omega_1 = -\omega$$

$$\frac{d}{ds}[\gamma e^{-\Omega_s/(2\pi i \hbar)}]_{2n} = d[-(2\pi i \hbar)^{-1} \gamma B e^{-\Omega_s/(2\pi i \hbar)}]_{2n-1}$$

FC8: γ is closed, its rank term is zero, and its positive-degree terms have compact support K.
The full primitive therefore has compact support; both characteristic integrals agree.

$$L \geq n + 1, \quad N \geq L + 1, \quad d_j = [\lambda^j](e_\infty - e_0), \quad a_\varepsilon(z) = a(\varepsilon z)$$

$$\text{ind } a^w = (2\pi)^{-n} \sum_{j=0}^{L-1} \varepsilon^{2j-2n} \int \text{tr}_{2\nu} d_j \Omega_{AN} + O(\varepsilon^{2L-2n})$$

AT1: both original ordered error powers, the full phase Jacobian and every Fourier factor remain.
Successive coefficient limits give zero for $j \neq n$ and the actual analytic index at $j = n$.

Proofs: FM6-FM10, FC8; RP12-RP15 at each finite degree, IP22-IP28, AT1.
Source framework: Pfäum-Posthuma-Tang, CyclicWeyl.tex 336-382;
central-lift observation: Feigin-Felder-Shoikhet, FFSrevised.tex 2189-2194.

Figure 1: The full central connection change and the original analytic coefficient comparison

and write $b = \chi a^{-1}$ there, extended by zero. Then $ba = ab = \chi I_\nu$, with the two matrix orders retained. Let ∂B_R , $R > R_0$, have the outward orientation induced by Ω_{AN} . The *radial profile* ψ_{rad} below is a different scalar function from χ .

Equations (RC3) and (RC9) construct, for every fixed sufficiently small $\varepsilon > 0$,

$$F_\varepsilon(z) = \psi_{\text{rad}}(\varepsilon z)z, \quad a_\varepsilon = a \circ F_\varepsilon \in S(1, g), \quad F_\varepsilon(z) = z \quad \text{when } |\varepsilon z| \leq 1. \quad (\text{PM2})$$

For $|\varepsilon z| \geq 2$, $F_\varepsilon(z) = z/(\varepsilon|z|)$, so its image is on the sphere of radius $1/\varepsilon$. Choose $1/\varepsilon > R_0$; then a_ε is uniformly invertible at phase infinity with the same exterior inverse bound as a , evaluated on that sphere. This checks every hypothesis of MB1–MB10 for a_ε ; it does not pretend that the endpoint $a = a_0$ lies in $S(1, g)$.

5.2 The fixed boundary and the analytic endpoint

Equations (PT2)–(PT9) construct both product-metric Weyl error sides for a_ε and $a_0 = a$, with the exact common compact cutoff defect $1 - \chi_\varepsilon = 1 - \chi$ for small ε . Taking $N = 2n + 2$, Equation (PT7) supplies one integrable majorant for every original weighted trace-class seminorm of both N -th error powers. Equation (PT9) then proves trace-norm convergence of those powers, and Equation (PT10) concludes

$$\text{ind}(a \circ F_\varepsilon)^w = \text{ind } a^w \quad \text{for all sufficiently small } \varepsilon > 0. \quad (\text{PM3})$$

This is an operator-index equality derived from the original full errors, not a claim of operator-norm convergence of a_ε^w .

After fixing $R > R_0$, make ε smaller so that $\varepsilon R < 1$. Then F_ε is the identity on a neighborhood of ∂B_R . Equation (PT11) gives both the exact symbol and tangential one-form equalities there:

$$a_\varepsilon|_{\partial B_R} = a|_{\partial B_R}, \quad (a_\varepsilon^{-1} da_\varepsilon)|_{T\partial B_R} = (a^{-1} da)|_{T\partial B_R}. \quad (\text{PM4})$$

The sphere, its outward orientation and all $2n - 1$ matrix-valued one-form factors are identical on the two sides. No global change-of-variables factor is needed: the comparison occurs on the **same fixed sphere**, where the radial map is literally the identity.

Apply the proved isotropic theorem MB10 to a_ε , then use (PM3)–(PM4). The exact product-metric matrix Weyl index is

$$\text{ind } a^w = -(-2\pi i)^{-n} \frac{(n-1)!}{(2n-1)!} \int_{\partial B_R}^{\Omega_{AN}} \text{tr}_\nu[(a^{-1} da)^{2n-1}]. \quad (\text{PM5})$$

The coefficient is MB10's original one, including its sign and both factorials. The proof holds at every $n \geq 1$ and matrix rank ν under the stated product-metric hypotheses.

5.3 The cutoff and left-quantization versions

The same χ in (PM1) gives $b = \chi a^{-1}$ and the full original compact differential form $(1 - \chi)^n \text{tr}[(db \wedge da)^n]$. Equations (BN1)–(BN5) derive its exact transgression and beta moment solely from this identity on the invertibility region and compact cutoff support. Applying (BN5) to the original a, b, χ , and comparing its written coefficient with (PM5), gives the equivalent **product-metric cutoff formula**

$$\text{ind } a^w = (2\pi)^{-n} \frac{2}{i^n n!} \int_{\mathbb{R}_{x,\xi}^{2n}}^{\Omega_{AN}} (1 - \chi)^n \text{tr}_\nu[(db \wedge da)^n]. \quad (\text{PM6})$$

The cutoff formula here is for the *original* product symbol, not for a_ε ; it follows from the original transgression and the common boundary integral. Every coefficient is the same as MB9 and equation (17) of From ordered Weyl errors to the exterior index form.

Finally (PT12) is the exact left-to-Weyl symbol identity

$$\text{Op}_0(a) = \text{Op}_{1/2}(a_W), \quad a_W = \exp\left(-\frac{i}{2}\langle D_x, D_\xi \rangle\right) a, \quad a_W - a \in S(h_G, G), \quad h_G = \frac{1}{\langle x \rangle \langle \xi \rangle}. \quad (\text{PM7})$$

The product Planck weight tends to zero at full phase infinity, so the actual operator difference is compact by the metric operator bounds lesson. Equation (PT14) proves equality of Fredholm indices. Equations (PM5)–(PM6) therefore give **the same exact numerical values** for the original left quantization:

$$\text{ind Op}_0(a) = \text{ind } a^w = -(-2\pi i)^{-n} \frac{(n-1)!}{(2n-1)!} \int_{\partial B_R}^{\Omega_{\text{AN}}} \text{tr}_\nu[(a^{-1} da)^{2n-1}]. \quad (\text{PM8})$$

The left and Weyl operators and their symbols remain distinct. Their proved compact correction identifies their indices. The full finite term-by-term integration is proved independently in From ordered Weyl errors to the exterior index form, equations (5)–(18).

The metric assumptions set the scope of this result. The isotropic metric is covered by the matrix-index argument, and the product metric by the radial transport just proved. For a general σ -temperate metric g , let g^σ be its symplectic dual. The further condition $\sup_{v \neq 0} g_z(v)/g_z^\sigma(v) \rightarrow 0$ as $|z| \rightarrow \infty$ does not by itself enter either proof above. This broader metric setting is not established by either proof above. No general-metric index theorem is asserted in this lesson.

6. Worked example: a nonzero scalar index

In one phase plane take

$$a(x, \xi) = \frac{x + i\xi}{\sqrt{1 + x^2 + \xi^2}}, \quad (x, \xi) \in \mathbb{R}^2. \quad (\text{MI1})$$

For every multiindex α , differentiating the numerator and denominator gives $|\partial^\alpha a(z)| \leq C_\alpha \langle z \rangle^{-|\alpha|}$; thus $a \in S(1, g)$. Its only zero is at the origin and $|a(z)| = |z|/\sqrt{1 + |z|^2}$ is bounded below outside any fixed ball about the origin. On the positively oriented circle $z = Re^{i\vartheta}$, the radial denominator is positive and $a^{-1} da = i d\vartheta$. Formula (MB10) at $n = 1$ has coefficient $-(-2\pi i)^{-1} = 1/(2\pi i)$. Therefore

$$\text{ind } a^w = \frac{1}{2\pi i} \int_{\partial B_R} a^{-1} da = \frac{1}{2\pi i} \int_0^{2\pi} i d\vartheta = 1. \quad (\text{MI2})$$

The full bounded symbol, the outward orientation and the original Weyl operator are used; no unbounded model is substituted.

7. Exercises with solutions

Exercise 1. Evaluate the exact boundary coefficient in (MB10) for $n = 2$, retaining its sign and factorials.

Solution. Since $(-2\pi i)^2 = -4\pi^2$ and $(n-1)!/(2n-1)! = 1/6$, the coefficient is $-(-4\pi^2)^{-1}/6 = 1/(24\pi^2)$. This uses the original $dx_1 \wedge d\xi_1 \wedge dx_2 \wedge d\xi_2$ orientation.

Exercise 2. Let $e_\infty - e_0 = \sum_{j \geq 0} \lambda^j d_j$ as in (MB11). Show what the full formal identity (MB7) says about d_{n+2} .

Solution. Its contribution to $\mathcal{T}_\lambda(e_\infty - e_0)$ is $(2\pi)^{-n}\lambda^2 \int \text{tr}_{2\nu} d_{n+2}$. The right side of (MB7) has no λ^2 term, so the integral is zero. No pointwise assertion about d_{n+2} follows.

References

The external higher algebraic index theorem is M. Pflaum, H. Posthuma and X. Tang, arXiv:0805.1411v3, original IndThms.tex, theorem thm:higher-algind. Its top local Riemann–Roch antecedent is B. Feigin, G. Felder and B. Shoikhet, arXiv:math/0311303v2, original FFSrevised.tex. These are cited source results; the specialization, exact normalizations and original-operator receiving argument are proved above. The direct term-by-term integration of the separate finite Weyl coefficient is completed in the direct exterior-reduction lesson, equations (5)–(18).

8. Editorial completion: a full proof of the operative flat pairing

The source statements, all formulas and their order above are retained. In this lesson (MB3) is an operative assertion: a citation alone would not prove (MB7)–(MB10). We now prove (MB3) for its exact written projectors and every dimension by the finite original Weyl calculation. This supplies the complete required theorem specialization within the programme. The more general theorem on other symplectic manifolds and higher closed sequences is not invoked to fill any step below.

8.1. Full tensor definitions and the actual smooth horizontal algebra

The exponential denoted by $\mathcal{E}$ in (WR9) is the complete source expression

$$\mathcal{E}(u) = \prod_{0 \leq r < s \leq 2n} \exp(\hbar(u_r - u_s + \tfrac{1}{2})\alpha_{rs}) \Big|_{u_0=0}, \quad \Delta^{2n} = \{0 \leq u_1 \leq \dots \leq u_{2n} \leq 1\}. \quad (\text{LP1})$$

Every pair, the $1/2$, the full parameter and the slot order are retained. Its scalar constant derivative operators commute; this permits regrouping derivative operators without regrouping the matrix factors. For matrix-valued slots, the map μ_{2n} evaluates their fiber arguments at zero and takes the ordinary trace of their product in the original order. Its scalar version omits only that final trace. A fixed formal coefficient contains finitely many input degrees and contraction degrees, so these maps are defined for smooth base coefficients as formal differential expressions.

For such a coefficient, the notation $u(p + P, q + Q)$ in (FM7) means its full formal Taylor jet

$$\hat{u} = \sum_{\alpha, \beta \in \mathbb{N}^n} \frac{P^\alpha Q^\beta}{\alpha! \beta!} (\partial_p^\alpha \partial_q^\beta u)(p, q). \quad (\text{LP2})$$

It asserts no convergence of a Taylor series of a smooth function. The equations $\partial_{P_j} \hat{u} = \partial_{p_j} \hat{u}$ and $\partial_{Q_j} \hat{u} = \partial_{q_j} \hat{u}$ follow by reindexing the individual coefficients, including the factorials. Conversely, these equations and the value at $P = Q = 0$ determine every coefficient by induction in its total fiber degree. Thus the horizontal-lift and evaluation maps are exact inverses. At every base formal degree, the diagonal chain rule and each finite fiber contraction give exactly (FM10). This proves the full smooth horizontal algebra and the exact (FM4) map with inverse, fixed supports and both matrix orders.

The determinant in (WR9) differentiates each of its $2n$ connection slots once. The full commutative exterior computation (WR4)–(WR6) proves the divided Pfaffian normalization (WR8): its coefficient of an ordered basis wedge is the sum of the distinct-plane determinants, each occurring once after the written division. Sorting the pair containing the first slot gives the full recurrence (WR3). Its base is $\Pi_0 = 1$, $\Pi_2 = \alpha_{01}$. Hence induction matches the source recurrence at every degree, rather than

assuming an exterior-power convention. On the actual A in (FM6), all its central contributions have zero fiber derivative and each remaining derivative is precisely $-dq_j$ or dp_j . Every determinant permutation therefore contributes the same oriented volume term in (FD6). After that operation every nonzero slot other than slot zero is fiber constant. Every positive term of each factor of (LP1) differentiates one of those constant slots, and is zero. There are no remaining exponential terms.

For completeness, the volume of $0 \leq u_1 \leq \dots \leq u_d \leq r$ is $r^d/d!$: it is one for $d = 0$, and integrating the $(d-1)$ -dimensional volume $v^{d-1}/(d-1)!$ over the final coordinate $0 \leq v \leq r$ proves the formula inductively. Thus all $(2n)!$ determinant terms cancel exactly the $1/(2n)!$ simplex volume, while the original $(-1)^n \hbar^n$ and $\hbar^{-2n}$ remain. This proves (WR12) and (WR13) for every finite matrix rank. At degree zero the pairing is evaluation on the actual relative difference; (NS2)–(NS4) have one summand with the complete $(2\pi i)^{-n}$ factor. Its cyclicity can also be proved here without a higher cocycle theorem: (WR13) is the constant multiple of the compact coefficient integral, and integration by parts annihilates every positive C_k by antisymmetry of J against symmetric second derivatives, as proved in FC1–FC3 of the relative-projector lesson. Thus its degree-zero trace and pairing have their actual domains and do not require an unproved general cyclic-cohomology assertion.

8.2. The full finite coefficient and every linear-path hypothesis

Fix an integer $N > n$ and retain the original $a, b, t = 1 - \psi$. Put $r_1 = I - b \#_{\lambda} a$, $r_2 = I - a \#_{\lambda} b$, with their full formal expansions. DE3–DE5 and OC27–OC30 of the scaled Weyl lesson construct the finite degree- n coefficient of $r_1^{\#N} - r_2^{\#N}$, with every intrinsic term and every later contraction. Both scaled analytic remainders lie in the original $S(\hbar_{\varepsilon}^{n+1}, g_{\varepsilon})$, with the complete product weight $(1 + \hbar_{\varepsilon}/4)^{4n}$ and its derivative comparison in RP1–RP5. RA1–RA10 prove their actual trace-norm bound $C\varepsilon^2$. Each retained coefficient is compact: in any degree at most $n < N$, one of the N intrinsic slots has degree zero, and its differentiated tI_{ν} is supported in the original K . CI12 gives its trace with $(2\pi)^{-n} \varepsilon^{-2n}$. T28 and IP16 give the index of the unchanged a^w at every positive scale. The finite asymptotic coefficient comparison consequently proves

$$\text{ind } a^w = (2\pi)^{-n} \int_{\mathbb{R}^{2n}} \text{tr}_{\nu}[\lambda^n](r_1^{\#N} - r_2^{\#N})(z) dz. \quad (\text{LP3})$$

To check that comparison directly, multiply the finite expansion by ε^{2n} and take its limit to prove the degree-zero coefficient zero; after each lower coefficient is zero, multiply by the power which makes the next coefficient constant and take its limit. At degree n the remainder tends to zero and gives (LP3). The whole calculation is finite and has a proved trace-norm remainder.

We will use arbitrary invertible real linear maps only through their exact original symbols. On a compact parameter interval let L_s be a continuously differentiable path of invertible matrices, with $\|L_s\| \leq C$, $\|L_s^{-1}\| \leq C'$. Then

$$c\langle z \rangle \leq \langle L_s z \rangle \leq C''\langle z \rangle, \quad \|\partial^{\gamma}(a \circ L_s)(z)\| \leq C_{\gamma}\langle z \rangle^{-|\gamma|}. \quad (\text{LP4})$$

The lower bound follows from $|L_s z| \geq \|L_s^{-1}\|^{-1}|z|$, retaining the term one in both brackets; the upper bound follows from the operator norm. The chain rule gives the second inequality as the complete finite sum of products of entries of L_s times the corresponding derivative of the unchanged a at $L_s z$. For its parameter derivative differentiate that same sum. A derivative falling on a matrix entry retains the same weight. A derivative falling on $a(L_s z)$ inserts $\dot{L}_s z$, bounded by $C|z|$, and one further derivative of a , with weight $\langle L_s z \rangle^{-|\gamma|-1}$; the extra $|z|$ leaves exactly the weight in (LP4). Integrating this bound in s proves continuity of every original symbol seminorm. B26 proves operator-norm continuity. Exterior invertibility is preserved because L_s^{-1} is uniformly bounded, and the original bounded inverse evaluates at $L_s z$. The corresponding cutoff is $\psi \circ L_s$, supported in the actual transformed inverse region, with a fixed compact support for its defect along this compact path. The full parametrix proves Fredholmness

at every s , so F6–F7 make the index constant along the path. No assertion about a non-symplectic Weyl algebra automorphism has been made.

Expand the integrand in (LP3) completely into occurrences of the original atomic factors a, b, t . Each term has exactly $2n$ labeled coordinate derivatives. Let I_α be the integral of the entire group with counts α_j in each original coordinate, including all its scalar coefficients and ordered matrix words, where $|\alpha| = 2n$. Positive diagonal maps D_ρ are joined to the identity by their positive diagonal path; (LP4) proves every hypothesis of the preceding index argument. Each derivative supplies its ρ_j , while the actual positive Jacobian is $\prod_j \rho_j$. Hence

$$\text{ind } a^w = (2\pi)^{-n} \sum_{|\alpha|=2n} I_\alpha \prod_{j=1}^{2n} \rho_j^{\alpha_j-1} \quad (\rho_1, \dots, \rho_{2n} > 0). \quad (\text{LP5})$$

This is a finite Laurent identity. Here is its complete independence argument. Include the exponent zero in its finite exponent set, and choose an integer vector v with pairwise distinct dot products on that set. Such a vector exists: $v = (1, m, \dots, m^{2n-1})$ fails a prescribed distinct pair only when the integer m is a root of its nonzero difference polynomial. Each such polynomial has finitely many roots by induction on its degree using polynomial division at a root, and there are finitely many pairs. Choose an integer outside their union. Substitution $\rho_j = e^{sv_j}$ gives a finite exponential identity. Differentiating it at $s = 0$ through one less than the number of exponents gives its Vandermonde system. The determinant is $\prod_{r < t} (c_t - c_r) \neq 0$: subtracting two equal columns at $c_r = c_t$ shows all these factors divide the determinant, and its highest-degree term fixes their coefficient to one. Thus the distinct coefficients are zero. Consequently

$$I_\alpha = 0 \quad (\alpha \neq \mathbf{1}), \quad \text{ind } a^w = (2\pi)^{-n} I_1. \quad (\text{LP6})$$

A reflection in one original coordinate multiplies the unique surviving derivative-count group by minus one; its absolute Jacobian is one. Applying (LP3)–(LP6) to that reflected symbol proves its index is the negative of the original. Every positive-determinant invertible matrix is joined to the identity by a path as in (LP4). To prove this last statement, Gram–Schmidt factors it into an orthogonal matrix and an upper triangular matrix with positive diagonal. Linear interpolation of the latter to its diagonal and then of its diagonal to the identity keeps every diagonal positive. An orthogonal matrix of determinant one is reduced to the identity by coordinate-plane rotations. For two entries a, b in slots $1, j$, if $r = \sqrt{a^2 + b^2} > 0$, the rotation with matrix $\begin{pmatrix} a/r & b/r \\ -b/r & a/r \end{pmatrix}$ sends them to $(r, 0)$; if both entries are zero, leave them unchanged. Applying this successively for $j = 2, \dots, 2n$ sends its first unit column to the first basis vector. Repeat on the orthogonal complement. The last one-dimensional orthogonal factor is one because the determinant remains one. Every rotation has its sine/cosine path from the identity, including the half-turn when $a < 0, b = 0$. The resulting finite paths remain invertible on compact intervals, so the bounds in (LP4) hold. For a coordinate permutation, its sign is therefore its index transport sign. This proves the complete alternation step in all dimensions.

8.3. The atomic involution, the surviving words, and the full count

Let S^1 be the original untraced differential expression with each coordinate count one, so that $I_1 = \int \text{tr } S^1$. For each coordinate-label permutation σ , apply (LP6) to $a \circ P_\sigma$ and change back to the unchanged integration variables. Its sign from the preceding paragraph is $\text{sgn } \sigma$. Thus each integral of $\text{sgn } \sigma S_\sigma^1$ equals I_1 , and

$$\text{ind } a^w = \frac{(2\pi)^{-n}}{(2n)!} \int \text{tr } S^2 dz, \quad S^2 = \sum_{\sigma \in S_{2n}} \text{sgn } \sigma S_\sigma^1. \quad (\text{LP7})$$

In each fully expanded atomic word, if two derivative slots reach the same occurrence of a, b or t , transpose their coordinate labels. The unchanged mixed derivatives commute on that occurrence; all other slots and every matrix position remain unchanged. The transposition reverses the alternating sign and has no fixed point because the labels are distinct. Pairing all permutations by this involution cancels that entire word. This is a finite sum of pairs, not an assumed matrix cancellation.

For completeness, every later contraction of an ordered N -fold product is generated by the finite degree part of

$$\exp \left(\frac{\lambda}{2i} \sum_{1 \leq r < s \leq N} \sum_{u,v} J^{uv} \partial_{z_{r,u}} \partial_{z_{s,v}} \right) \prod_{r=1}^N f_r(z_r) \Big|_{z_1 = \dots = z_N = z}. \quad (\text{LP8})$$

Its coefficients follow by induction from the diagonal chain rule: contracting a product of the first slots against the next slot is the sum of their pair contractions. These scalar operators commute, so the finite multinomial formula gives the displayed exponential coefficients, with their original factorials. Matrix factors keep their order throughout. Any intrinsic $C_k(b, a)$ or $C_k(a, b)$ with $k \geq 2$ has two derivatives on each of its two atomic occurrences, and cancels by the involution. Any later contraction reaching a first intrinsic bracket likewise makes one of its atomic occurrences carry a second derivative, and cancels. Among remaining scalar t -occurrences, the contraction graph has degree at most one at every vertex, and hence is a matching. Each of its edges is the exact scalar factor $\sum_j (t_{\xi_j} t_{x_j} - t_{x_j} t_{\xi_j}) = 0$. The scalar factors can be moved past the other factors without moving any matrix factor; that whole edge contribution is zero. These cases exhaust every contraction in (LP8).

Only undifferentiated scalar t 's and first intrinsic brackets remain. Since $-C_1(b, a) = (i/2)\{b, a\}$, their full expression before taking the alternating sum is

$$\binom{N}{n} t^{N-n} \left(\frac{i}{2} \right)^n (\{b, a\}^n - \{a, b\}^n), \quad \{b, a\} = \sum_j (b_{\xi_j} a_{x_j} - b_{x_j} a_{\xi_j}). \quad (\text{LP9})$$

Exactly n of the N original error slots supply a bracket, accounting for the full binomial. The two matrix brackets have not been replaced by negatives of one another.

Put $\tilde{\Omega} = d\xi_1 \wedge dx_1 \wedge \dots \wedge d\xi_n \wedge dx_n = (-1)^n \Omega_{\text{AN}}$. A bracket word using each coordinate once uses each canonical pair once. It has $n!$ choices of the order of pairs and 2^n choices of their internal bracket terms. For any fixed ordered exterior word in $(db \wedge da)^n$, every one of these $2^n n!$ base words has one unique coordinate-label permutation taking it to the target word. The base bracket sign times that permutation sign is precisely the target exterior sign: an internal pair switch contributes one minus, and a switch of two length-two blocks contributes plus. Thus every exterior word occurs with exactly the same multiplicity. The original factor $(i/2)^n$ leaves $n!i^n$, proving the untraced identity

$$S^2 \tilde{\Omega} = n!i^n \binom{N}{n} t^{N-n} ((db \wedge da)^n - (da \wedge db)^n). \quad (\text{LP10})$$

The finite graded matrix trace is proved entry by entry in ES1 of the relative-projector lesson. Moving the first one-form past the other $2n - 1$ one-forms gives $\text{tr}(da \wedge db)^n = -\text{tr}(db \wedge da)^n$. Inserting (LP10) in (LP7), with the pair-order sign and every factorial retained, gives

$$\text{ind } a^w = (2\pi)^{-n} \frac{2n!}{i^n (2n)!} \binom{N}{n} \int^{\Omega_{\text{AN}}} t^{N-n} \text{tr}_\nu[(db \wedge da)^n]. \quad (\text{LP11})$$

The numerator is twice $n!$, and the denominator retains $(2n)!$. Choosing $N = 2n > n$ retains t^n and the exact factor $\binom{2n}{n}$, so

$$\text{ind } a^w = (2\pi)^{-n} \frac{2}{i^n n!} \int^{\Omega_{\text{AN}}} (1 - \psi)^n \text{tr}_\nu[(db \wedge da)^n]. \quad (\text{LP12})$$

This proves (MB9) directly at its entire original scope. Every finite original Weyl term was either retained in (LP9) or paired by an explicit zero mechanism before this integration. Sections 2–6 of the direct exterior-reduction lesson record the same calculation; the detailed proof above supplies its use within the present lesson without importing an external index theorem.

8.4. The exact two sides of MB3, including the full curvature

The relative-projector lesson's FC5–FC11 prove, by arbitrarily deep finite expansions and actual trace-norm remainders,

$$\mathcal{T}_\lambda(e_\infty - e_0) = (2\pi\lambda)^{-n} \tau_{2\nu}(e_\infty - e_0) = \text{ind } a^w. \quad (\text{LP13})$$

Its complete proof is also the finite argument already written in (AT1): for each prescribed coefficient choose a fixed deeper integer truncation before sending ε to zero. It does not assign convergence to a formal series. Combining the actual flat density (WR14), (LP13) and (LP12) gives its entire source pairing value $(-1)^n$ times the right side of (LP12).

Here are all characteristic-side prerequisites with no imported theorem. The global conjugating matrix U_0 supplies a smooth frame for V_P from its first ν columns and a complementary frame from its other columns, so both the image and kernel of the actual nonorthogonal P are smooth bundles. The original Grassmann operator Pd obeys the connection product rule; its full curvature calculation is (CN4). Differentiating $P^2 = P$ makes dP off-diagonal, so $(dP)^2$ commutes with P . This proves (CN5), and the finite trace of every odd power is zero by that block decomposition. Thus all the closed relative forms in (FC3) are globally defined and compact; their degree-zero rank difference is identically zero. The chosen tangent connection has zero curvature, so the complete formal series $\det[(Y/2)/\sinh(Y/2)]^{1/2}$ has value one at $Y = 0$, its constant term. No statement about a nonflat characteristic class is needed.

The tangent connection choice also has a complete local comparison. Any other smooth connection on this globally trivial tangent bundle has a matrix one-form C . Put $C_s = sC$ and $R_s = dC_s + C_s \wedge C_s$. Direct expansion proves $d_{C_s} R_s = 0$ and $\partial_s R_s = d_{C_s} C$, where $d_{C_s} v = dv + C_s \wedge v - (-1)^{\deg v} v \wedge C_s$. The full graded matrix trace therefore gives $d \text{tr } R_s^j = 0$ and $\partial_s \text{tr } R_s^j = j d \text{tr}(C \wedge R_s^{j-1})$, for every $j \geq 1$, with no reordered matrix products. At each exterior degree the written characteristic series $\det[(R_s/2)/\sinh(R_s/2)]^{1/2}$ is a finite polynomial in these traces. Indeed the formal identity $\det f(Y)^{1/2} = \exp[\frac{1}{2} \text{tr} \log f(Y)]$, with $f(Y) = (Y/2)/\sinh(Y/2)$ and constant term one, follows by the adjugate differentiation formula for the determinant, its inverse geometric series, and equality of the initial constant terms; comparison of successive formal coefficients proves the identity. Thus each positive-degree derivative of the characteristic form is exact: apply the displayed trace derivative to one factor of each finite polynomial and retain the other closed factors. Integrating in s gives $\hat{A}(C) - 1 = d\eta_C$ with the full characteristic coefficients retained. The product of η_C with the original closed $\gamma \wedge e^{-\Omega_F/(2\pi i \hbar)}$ is compact. The proved compact Stokes receiver consequently shows that this actual connection change leaves the entire relative characteristic integral unchanged coefficient by coefficient. This establishes the connection-choice assertion in Section 3 without invoking a general characteristic-class theorem.

The radial primitive (FC5) also has an exact coordinate proof. Write $\beta_{ij} = -\beta_{ji}$ for a closed two-form and $(H\beta)_j = \sum_l z_l \int_0^1 s \beta_{lj}(sz) ds$. Differentiating under this compact parameter integral and using the full closedness identity gives

$$\begin{aligned} (dH\beta)_{ij} &= \int_0^1 \left[2s \beta_{ij}(sz) + s^2 \sum_l z_l \partial_l \beta_{ij}(sz) \right] ds \\ &= \int_0^1 \frac{d}{ds} (s^2 \beta_{ij}(sz)) ds = \beta_{ij}(z). \end{aligned} \quad (\text{LP14})$$

The endpoint at zero is zero by smoothness. Hence every original formal coefficient of Ω_F has its written global primitive. In top degree, the full exponential in (CN1) is a finite sum of exterior powers; a prescribed Laurent coefficient of it is a finite sum as well because its exterior power is bounded by n and the curvature has no negative intrinsic formal powers. Each term with a positive curvature power has exactly the compact primitive (FC6), including its factorial, power of $2\pi i$ and negative parameter powers. ES5–ES7 of the relative-projector lesson prove the required full-space and ball Stokes receivers in the actual original orientation. Their integral is therefore zero; no curvature contribution was suppressed before this exact comparison.

The surviving characteristic term is the full Grassmann top form $1/[n!(2\pi i)^n] \text{tr } P(dP)^{2n}$. The original compact exact defect (CP11)–(CP14), with its full polynomial and both endpoints, proves its integral equals the right side of (LP12). The source oriented coordinate swap supplies the single $(-1)^n$ in (CN7). Thus its characteristic value is also $(-1)^n$ times that same right side. We have proved both sides of (MB3) equal, on its exact actual pair, all matrices and all dimensions. This proves (MB3), (MB7), (MB9) and their formal and analytic receiving maps without assuming the higher algebraic index theorem or local Riemann–Roch theorem. Their cited general formulations remain free reference context; they are not missing proof prerequisites for these original programme conclusions.

9. Editorial strengthening: every central formal lift and every invertible linear map

9.1. Arbitrary higher curvature terms are actually realized

Keep the same unchanged flat horizontal algebra, and let β_j be any closed smooth scalar two-forms for $j \geq 1$. Define the full central one-form and curvature

$$B_\beta = -2 \sum_j p_j dq_j + \sum_{r \geq 1} \hbar^r H \beta_r, \quad A_\beta = A_0 + B_\beta, \quad \Omega_\beta = -\omega + \sum_{r \geq 1} \hbar^r \beta_r. \quad (\text{CL1})$$

The commutator with a central scalar one-form is zero as a derivation of sections, so (FM7) and the horizontal jets (LP2) are unchanged. Its graded self-commutator and cross commutators are zero because exterior scalar one-forms anticommute and scalar coefficients commute. The full curvature is therefore $dB_\beta + \hbar^{-1} A_0 \wedge_\star A_0 = -\omega + \sum \hbar^r \beta_r$, by (FM8) and (LP14). This proves an actual connection realizing every written higher coefficient, rather than merely an exactness assertion about a prospective curvature.

Every term of the determinant density containing any slot from B_β is zero after that slot's fiber derivative. Thus the full density and its degree-zero pairing are identical coefficient by coefficient for all these lifts. On the characteristic side, the full transgression (FC8) with $B_\beta - B$ has compact primitive, retaining all exponential factors and inverse powers; equivalently (FC6) proves all positive curvature terms integrate to zero. Hence the completely proved equality (MB3) holds for the exact same relative pair under every lift (CL1). The identity map on horizontal sections, the identical algebra product, the unchanged compact support and the explicit central curvature morphism are all proved. No general existence theorem for nonflat Fedosov connections is being used.

9.2. Linear index transport and the exact algebra defect

Let $L \in GL(2n, \mathbb{R})$, preserve $a_L = a \circ L$, $b_L = b \circ L$, $\psi_L = \psi \circ L$, and $K_L = L^{-1}K$. The exact path proof (LP4) and its reflection computation give the stronger index comparison

$$\text{ind}(a \circ L)^w = \text{sgn}(\det L) \text{ind } a^w. \quad (\text{LI1})$$

For negative determinant, compose with one fixed coordinate reflection to obtain positive determinant, apply its proven path and then the negative sign. All transformed original symbols remain in the same isotropic class with their own exact finite bounds; no quantization identity for a general linear change

was presumed. Applying (LP13) to each rebuilt original relative projector gives the corresponding exact integrated formal trace with that same sign. This is index transport between the actual bounded operators, not equality of their kernels.

There is also a full algebra map, with its exact failure object in the original product. Write $J_L = L J L^T$, retain all original ordered derivatives, and define $\#_\lambda^{J_L}$ by the complete formula (RP7) with this displayed tensor. The chain rule proves at every coefficient

$$(f \circ L) \#_\lambda^{J_L} (g \circ L) = (f \#_\lambda^{J_L} g) \circ L. \quad (\text{LI2})$$

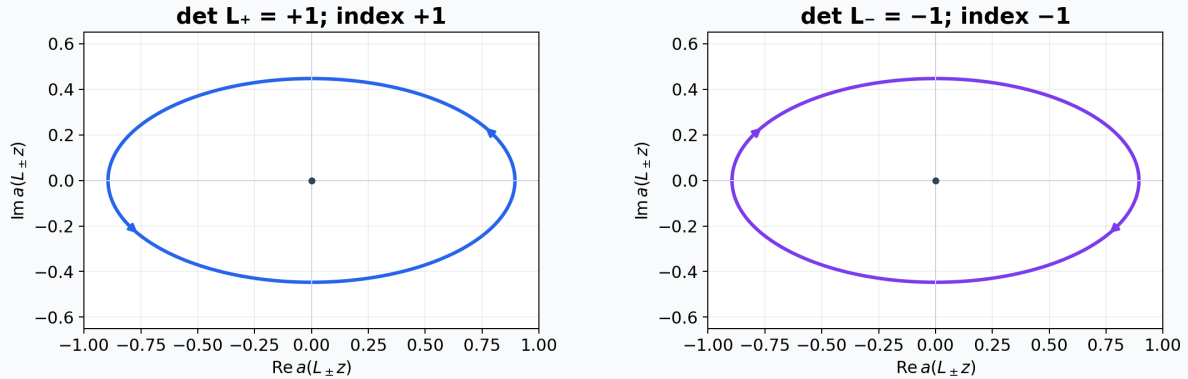
This defines an exact algebra isomorphism from the smooth formal algebra with tensor J_L to that with tensor J , with inverse pullback by L^{-1} ; supports map exactly by L^{-1} , and matrices retain their order. In the unchanged original J -product its defect is the fully specified bilinear map

$$\begin{aligned} \mathcal{D}_L(f, g) &:= (f \circ L) \#_\lambda^J (g \circ L) - (f \#_\lambda^J g) \circ L \\ &= \sum_{k \geq 1} \frac{\lambda^k}{(2i)^k k!} \left[\prod_{r=1}^k (J_L)^{u_r v_r} - \prod_{r=1}^k J^{u_r v_r} \right] [(\partial_{u_1} \dots \partial_{u_k} f)(\partial_{v_1} \dots \partial_{v_k} g)] \circ L. \end{aligned} \quad (\text{LI3})$$

For formal input series the displayed contraction is further summed over their finite intrinsic degrees, as in (RP7). Thus the nonzero defect defines an exact family of deformed algebra structures and the actual morphism (LI2); it is not an assertion of unrelated objects. If $J_L = J$, the defect is zero at every order. If $J_L = -J$, the same full comparison is the parameter map $\lambda \mapsto -\lambda$, retaining its $(-1)^k$ in every degree. A nonsymplectic index comparison (LI1) therefore does not silently replace a product map by a coordinate-only automorphism.

Linear index transport retains the full original symbol

$$a(z) = (x + i\xi)/\sqrt{1 + x^2 + \xi^2}, \quad L_+ = \text{diag}(2, 1/2), \quad L_- = \text{diag}(-2, 1/2)$$



$$z = (\cos \theta, \sin \theta), \quad a(L_\pm z) = \frac{\pm 2 \cos \theta + (i/2) \sin \theta}{\sqrt{1 + 4 \cos^2 \theta + (1/4) \sin^2 \theta}}$$

$$J_L = L J L^T: \quad J_{L_+} = J, \quad J_{L_-} = -J; \quad \lambda \mapsto \lambda \text{ or } -\lambda$$

LI2 proves the full product morphism; LI3 retains every term of its defect in the original product.

LI1 proves the index sign for every invertible real L by the actual norm-continuous Fredholm paths.

Exact 2D symbol-image samples of the original positively oriented unit circle. Arrows follow increasing θ .

Complete proofs: LP3-LP12, LI1-LI3 and MI1-MI2. Reproducible programme calculation; no novelty claim.

Figure 2: Exact linear pullbacks of the original bounded one-plane symbol

These are exact two-dimensional symbol images of the original positively oriented unit circle under the full MI1 symbol composed with $L_+ = \text{diag}(2, 1/2)$ and $L_- = \text{diag}(-2, 1/2)$. The complete denominator

remains displayed; the arrows follow increasing circle parameter. LI1 proves the respective original bounded Weyl indices +1 and -1. LI2–LI3 prove the exact tensor and parameter comparisons, rather than infer them from the pictures. The reproducible figure source accompanies the full proof.

10. Editorial proof of the product endpoint and the original example

10.1. Every operative endpoint bound and both quantizations

For (PM1)–(PM8), use the exact radial profile and separate cutoff already defined there. RC4–RC7 of the radial transport lesson prove all separate-coordinate bounds of the full composition. To spell out their receiver, $\varepsilon\langle x \rangle$ and $\varepsilon\langle \xi \rangle$ are each at most $\langle \varepsilon z \rangle$. Distributing any mixed derivative of the profile yields the complete bound $C\psi(\varepsilon z)\langle x \rangle^{-|\alpha|}\langle \xi \rangle^{-|\beta|}$. Every chain-rule term of $a \circ F_\varepsilon$ contains exactly as many positive powers of $\langle F_{\varepsilon,x} \rangle$ and $\langle F_{\varepsilon,\xi} \rangle$ from the derivatives of the two components as the derivatives of the unchanged a at that point contain negative powers. They cancel term by term, leaving the stated original G -seminorm with uniform constants. On the outer region for fixed positive ε , the map $z/(\varepsilon|z|)$ has compact image and its derivative of order k has degree $-k$, so the original compact part and outer part give the exact $S(1, g)$ estimate separately. None is claimed uniform in that isotropic class at zero.

Let $c = \min_{r \geq 1} r\psi_{\text{rad}}(r) > 0$. Choose $0 < \varepsilon_* \leq 1$ with $\varepsilon_* R_0 < \min(1, c)$. On $|\varepsilon z| \leq 1$ the compression is the identity; on its complement $|F_\varepsilon(z)| \geq c/\varepsilon > R_0$. These two actual domains prove $1 - \chi \circ F_\varepsilon = 1 - \chi$ globally. The compact zeroth defect has one fixed support; its original positive h_G has a positive minimum there, so it lies in every fixed $S(h_G^q, G)$.

Both full ordered errors are uniformly in $S(h_G, G)$. At each product the complete provider weight is $(1 + h_G/4)^{4n} \leq (5/4)^{4n}$; WG4 retains its 4^{-N} , directional $2^{k/2}$ and tensor 2^J factors in every derivative estimate. Since symbol seminorms divide by the weight rather than differentiate it, this pointwise bound gives the exact target inclusion with that full constant. Iterating a fixed finite number of products retains the product of these constants, proving every ordered error power in $S(h_G^N, G)$. These symbols and the input $S(1, G)$ symbols have bounded Euclidean derivatives of every order, because the separate coordinate weights are at most one. The complete Schwartz/operator composition WO6–WO8 consequently applies in the Euclidean order-zero class; B26 then extends its identities to bounded maps on the original L^2 spaces by density. This proves the actual PT5–PT6 domains.

For the original integer $N = 2n + 2$, every actual CT1 term with total monomial and derivative degree at most $n + 1$ has the full bound

$$C\langle x \rangle^{|\alpha| - N - |\alpha'|} \langle \xi \rangle^{|\beta| - N - |\beta'|} \leq C\langle x \rangle^{-(n+1)} \langle \xi \rangle^{-(n+1)}. \quad (\text{PE1})$$

Its square is integrable separately in both n -dimensional coordinate groups: on dyadic annuli of radius 2^j , the bound is a geometric series with exponent $n - 2(n + 1) < 0$, and the compact ball has finite volume. Exact local smooth continuity of the complete product sends the full errors and their powers to their endpoint symbols. Dominated convergence applies to each of the finitely many original weighted CT1 terms, using twice the common bound in (PE1). CT2 gives trace-norm convergence of both error powers. T28 expresses each actual Fredholm index as their trace difference; trace continuity gives convergence to the endpoint index. Since both values are integers, choose an endpoint neighborhood in which their difference has absolute value less than one to obtain exact equality. This proves (PM3) with the actual original operators, without an operator-norm convergence claim for that radial endpoint.

For any fixed original $R > R_0$, choose $\varepsilon R < 1$. The compression is literally the identity on a neighborhood of the unchanged sphere, so its tangent map and all $2n - 1$ factors are identical, proving (PM4). The already proved (LP12) and the full cutoff primitive of BN1–BN5 yield (PM5)–(PM6) with the original sign and factorials. For left quantization, W40 gives the full exponential multiplier in (PM7), including $-i/2$. In its original product metric put $c = -1/2$, $A_c(p, q) = cp \cdot q$ and $B_c = (c/2) \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$. Direct inversion

gives $G^{A_c} = 4G^\sigma/c^2$, so its full phase parameter is $|c|h_G/2 = h_G/4$. The temperateness comparison is exactly $1 + G^\sigma = 1 + (c^2/4)G^{A_c}$, with the same actual displacement in both terms. The finite-bound Gauss remainder G24–G26, applied to this phase on the full $2n$ -dimensional space, retains the counting factor $(1 + |c|/2)^{2n} = (5/4)^{2n}$, all original structural constants, and its first remainder factor $h_G/4$. Its full directional bound is $C_{1,l,1/4}(h_G/4) \prod_{j=1}^l G(T_j)^{1/2} p_{\leq J}(a; 1, G)$. Consequently $a_W - a \in S(h_G, G)$ with the displayed factor $1/4$ retained in the seminorm estimate. B35 proves its finite-matrix Weyl operator compact: the full identity $\langle x \rangle^2 \langle \xi \rangle^2 = 1 + |x|^2 + |\xi|^2 + |x|^2 |\xi|^2 \geq \langle z \rangle^2$ gives $h_G \leq \langle z \rangle^{-1} \rightarrow 0$ in full phase infinity. F10 supplies the exact compact Fredholm stability, with all operators on the same original L^2 domain and target, so the two original indices agree while their operator difference and symbols stay explicit. This closes all of (PM1)–(PM8) at their written original scope.

10.2. Every derivative and the actual nonzero example

Put $r(z) = 1 + |z|^2$. Repeated differentiation of $r^{-1/2}$ gives a finite sum of $c z^\delta r^{-1/2-j}$ with $2j - |\delta| = |\alpha|$: differentiating its polynomial part decreases $|\delta|$ by one, and differentiating its denominator increases j by one and $|\delta|$ by one. Both preserve that equation at the next derivative degree. Hence every term is bounded by $C \langle z \rangle^{-1-|\alpha|}$. The complete product rule for the original degree-one numerator $x + i\xi$ has only its zero and first derivative terms. Multiplication of the first bound by $|z|$, or use of the bound with one fewer denominator derivative for the second term, gives precisely $C_\alpha \langle z \rangle^{-|\alpha|}$. This proves every original MI1 symbol seminorm, with its full denominator retained.

The original modulus $|z|/\sqrt{1 + |z|^2}$ has its only zero at zero, is increasing in $|z| > 0$ by differentiating the full quotient, and is bounded below outside any fixed positive-radius ball. On the original positively oriented circle $(x, \xi) = (R \cos \vartheta, R \sin \vartheta)$, the full symbol is $Re^{i\vartheta}/\sqrt{1 + R^2}$; its reciprocal differential is exactly $i d\vartheta$. The boundary theorem now has the complete proof (LP3)–(LP12) and the already proved global cutoff moment, so (MI2) is the index of this original bounded Weyl operator, equal to one. No unbounded operator model is substituted.

As further exact consequences, a direct sum of two symbols satisfying the same hypotheses has the sum of their indices: choose one cutoff supported in both inverse regions and use the ordinary finite block trace in (LP12) or (PM6). A globally uniformly invertible original symbol has index zero, by the following exact extension of the calculation to the identity cutoff. Repeatedly differentiating $a^{-1}a = I$ writes every derivative of a^{-1} as a finite ordered sum of inverse factors and derivatives of a ; their total derivative orders add to the original order. The uniform inverse bound therefore gives the same full symbol derivative estimates for $b = a^{-1}$. Set $\psi = 1$, so the unchanged zeroth defect is $tI = 0$, with empty compact support. All finite composition, trace-remainder and Fredholm hypotheses of (LP3) remain valid with this globally smooth b . In every coefficient of degree $n < N$, at least one of the N intrinsic error slots has degree zero and is identically zero, even after all its later derivatives. Thus (LP3) gives index zero. Equivalently the full classical conjugating matrix is $\text{diag}(a, a^{-1})$ and its relative projector is exactly e_0 . This extension removes the unnecessary inner cutoff only when the original inverse is globally defined and uniformly bounded; the earlier exterior-inverse proof and its cutoffs remain intact. It does not assert that the Weyl operator itself is invertible. Finally, in scalar rank $\nu = 1$ and $n \geq 2$, the scalar one-form $a^{-1}da$ squares to zero by exterior antisymmetry, so its $(2n - 1)$ -st power is zero. The exact boundary formula proves its index zero under either written metric, and also for the written left quantization. These are consequences for the original objects, with no hypothesis or factor removed.

11. Editorial proof of the finite factorization used in the linear path

Here is the exact finite construction used in Section 8.2. Let the original real invertible matrix L have columns $v_1, \dots, v_{2n}$. Define successively

$$w_j = v_j - \sum_{k < j} (q_k^T v_j) q_k, \quad r_{jj} = \sqrt{w_j^T w_j}, \quad q_j = w_j / r_{jj}, \quad r_{kj} = q_k^T v_j \ (k < j), \quad r_{kj} = 0 \ (k > j). \quad (\text{QR1})$$

The induction starts with $w_1 = v_1 \neq 0$. If the earlier q_k are orthonormal, then $q_k^T w_j = 0$ for every $k < j$. If $w_j = 0$, the formula writes v_j in the span of the earlier q_k , which by the earlier formulas is the span of $v_1, \dots, v_{j-1}$; this contradicts invertibility. Thus every displayed square root is strictly positive. The new q_j is unit and orthogonal to the earlier vectors. This proves the induction and, with the actual columns retained, $L = QR$, $Q^T Q = I$, and R upper triangular with its written positive diagonal.

For $\det L > 0$, the full determinant gives $\det Q = \det L / \prod_j r_{jj} > 0$. Orthogonality gives $(\det Q)^2 = 1$, so it is exactly one. Set $D = \text{diag}(r_{11}, \dots, r_{2n, 2n})$. The two actual paths

$$R_s = (1 - s)R + sD, \quad D_s = (1 - s)D + sI, \quad 0 \leq s \leq 1 \quad (\text{QR2})$$

retain respectively the original positive diagonal entries and the entries $(1 - s)r_{jj} + s > 0$. Their determinants are the products of these entries because the matrices are upper triangular. Consequently QR_s joins L to QD , and QD_s joins QD to Q , through invertible real matrices. The explicit coordinate-plane rotations already proved in Section 8.2 join Q to I . This gives the claimed finite piecewise differentiable path with every endpoint. On each compact segment the adjugate formula for the inverse and the strictly nonzero continuous determinant give the finite uniform bounds on L_s and L_s^{-1} used in (LP4). The actual symbol, cutoff, operator and index comparisons therefore have all their finite factorization prerequisites proved within this lesson.